



Artificial Intelligence and Machine Learning

S Thenmozhi

Department of Computer Applications

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AI Landscape

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Making the Data tidy

1. Removal of Duplicates

Duplicates are the records with same values except the change in the index numbers

```
#dataframe of duplicate values are made  
dup=df[df.duplicated()]
```

```
# delete the records which are duplicated  
df=df.drop(dup.index,axis=0)
```

2. Removal of Outliers

Data out of range is termed to be outliers. Outliers can be seen with the help of the boxplot.

If a datapoint is beyond $q1 - 1.5 * IQR$ or $q3 + 1.5 * IQR$, then it is identified as outlier. (IQR represent the interquartile range which is $(Q3 - Q1)$)

How to code it in Python?

```
Q1 = df['salary'].quantile(0.25)
```

```
Q3 = df['salary'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
# identify outliers
```

```
outliers = df[(df['salary'] < Q1 - 1.5 * IQR) | (df['salary'] > Q3 + 1.5 * IQR)]
```

```
# drop rows containing outliers
```

```
df = df.drop(outliers.index)
```

Note: You can also do it with beyond +/- 3 stddev also

3. Missing Value Treatment

Removing and imputing missing values from the dataset

Missing Values – blank values in the table

When you read the csv file the missing values is considered as NaN/NaT

Missing values will produce unpredictable results

So Treat the missing values....

Eliminate the samples or features with missing values

Can be done with dropna in python

Dropping rows (axis=0)

Variations – how=“all” / “any”

thresh=4 (keeps only records which has at least 4 non-NA values)

Dropping columns (axis=1)

Imputing missing values

Use interpolation techniques to estimate the missing values

Fillna() – Easiest way

Most popular: Mean imputation

Can be done using scikit-learn imputer helps to choose the strategy = “mean” or “median”

- axis=0 for row means

- axis=1 for column means

For categorical values – use “most_frequent”

Imputer

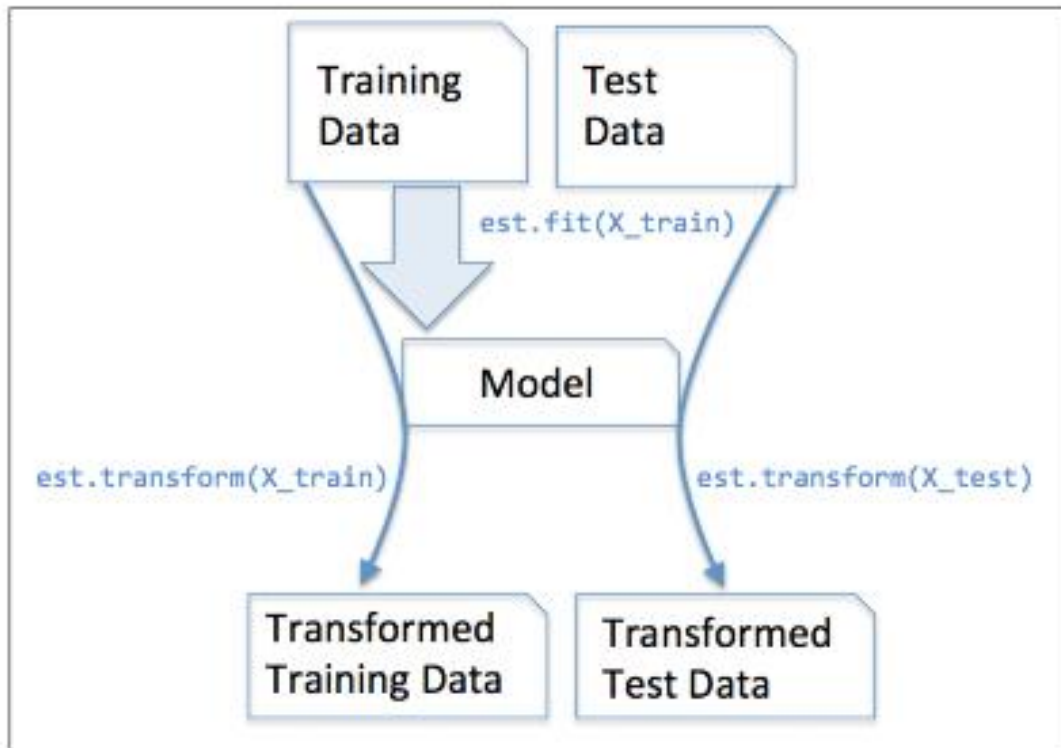
Belongs to Transformer class of scikit-learn

Two essential methods – **fit and transform**

Fit - Learn the parameters from the training data

Transform - Transform the data

The Number of features taken for fit and transform should be the same





THANK YOU

S Thenmozhi

Department of Computer Applications

thenmozhis@pes.edu

+91 80 6666 3333 Extn 393